Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18ME54]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

CONTROL ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

i. Answer **FIVE** full Questions. Question 1 and 2 are compulsory.
ii. Each question carries 14 marks.
iii. Missing data may be suitably assumed.

1. a. Determine the governing equations of mechanical system shown in fig. 1a. Also sketch Force-Voltage and Force current circuits using analogous quantities. **07**

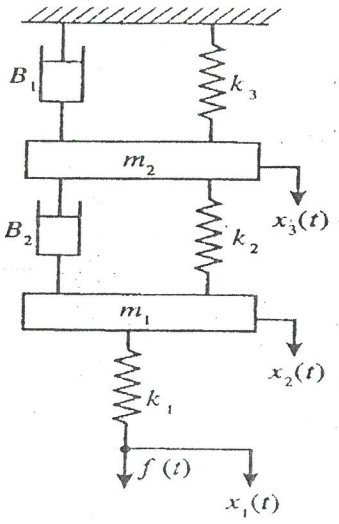

Fig. 1a

b. Define control system and distinguish between open loop and closed loop control systems. **07**

2. a. **07**

Reduce the block diagram shown in fig. 2a to canonical form and determine its transfer function.

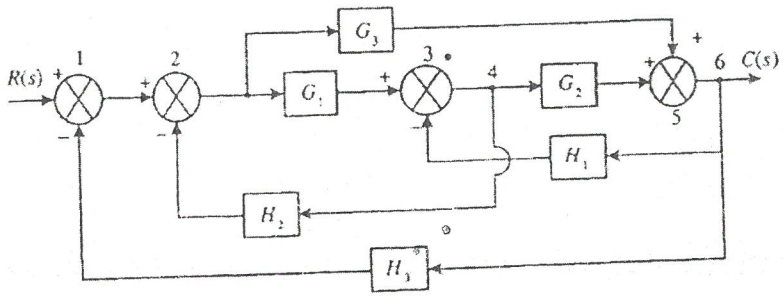

Fig. 2a

b. Using signal flow graph and Mason's gain formula obtain the overall transfer function of the system shown in fig. 2b. **07**

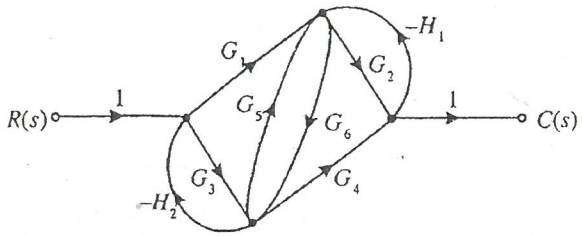

Fig. 2b

3. a. Define **08**

   i. Gain Margin

   ii. Phase Margin

   iii. Gain crossover frequency

   iv. d. Phase crossover frequency

b. Using Routh Hurwitz criterion ascertain the stability for the system **06**

$S^6 + 2S^5 + 8S^4 + 12S^3 + 20S^2 + 16^S + 16 = 0.$

OR

4. a. Sketch the root locus for negative feedback system whose open loop transfer function is given by. **14**

$$G(s)H(s) = \frac{K(s+1)}{s^2(s+3)(s+5)}$$

Determine the range of variation of K over which system remains stable.

5. Sketch the Bode plot for the transfer function **14**

$$G(s)H(s) = \frac{80}{s(s+2)(s+20)}$$

Obtain gain margin, phase margin and cross over frequencies, also determine the gain margin when frequency is 6.35 rad/s

OR

6. Sketch the bode plot for the transfer function **14**

$$G(s)H(s) = \frac{Ks^2}{(1+0.02s)(1+0.2s)}$$

Obtain gain margin, phase margin and cross over frequencies, also determine the value of K when gain crossover frequency is 5 rad/s

7. a. Derive transfer function for the lag compensator with circuit diagram along with its characteristics. **10**

b. Enumerate the differences between classical approach and modern approach **04**

OR

8.	**a.**	Derive transfer function for the lead compensator with circuit diagram along with its characteristics.	**10**
	b.	Define controllability and observability.	**04**